Short distribution representatives at fan-out two: audit of the relaxed-witness lemma and the abelian-filter theorem, Parikh rigidity of intra-word exchanges, and polynomial data for semilattice-split abelian unions

v80 research draft for #2 (fixed-h MCFG), Conjecture E

Research memo (internal)

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Abstract

We re-audit the two results of the v79 research draft (PR #46) on which Conjecture E rests: the relaxed-witness lemma (R) and the polynomial-data theorem for abelian-filtered targets (D). Both hold as stated; we record the exact dependencies, the uniformity of D in the auxiliary abelian group, and one interaction with the canonical v79 rule-only witness proposition. We then prove: (P) every arity-d exchange pair witnessed inside a single word is Parikh-equal, so the local mobility lemmas (B, B′) can never shorten a tuple; and (F1-sl) for every finite typing h refining the two-element semilattice typing h_{U}, every fan-out f and rank r, targets of the form $\{w \text{ marker-free} : \varphi(w) \in \gamma_1 + \mathcal{H}\} \cup \{w \text{ with a marker} : \varphi(w) \in \gamma_2 + \mathcal{H}\}$ have characteristic samples of size polynomial in $\|G\| \max\{1, \theta_G\}$, uniformly in the abelian data. These targets are outside the abelian-filter class whenever $\gamma_2 - \gamma_1 \notin \mathcal{H}$, and h_{U} is outside **LG**; hence F1-sl is a genuinely new instance of Conjecture E. The key new ingredient is an *offset-rigidity* lemma: under a marker-free context a good rule can have at most one child whose yields contain both marker-free and marked tuples. Conjecture E itself remains open.

1 Conventions

We use the v79 manuscript (`PAPER.yaml` internal version v79, HEAD `7ef2cb1`) verbatim: sentence contexts have nonempty internal separators; tuple components are nonempty; $\mathcal{D}_L(\vec{x})$ is the set of accepting sentence contexts of the arity of $\vec{x}$; $\vec{x} \equiv_{L,h} \vec{y}$ means equal componentwise h-types and equal distributions; $\mathcal{C}^{\mathrm{mcf}}_{f,h}$ is the class of (f,h)-tuple-substitutable f-MCFLs; $\mathcal{B}^{\le r}_{f,h}$ is the capped reconstruction operator. For a reduced good presentation $G = (V, \Sigma, P, S)$ let $\widetilde{G}_0$ be its trimmed yield-type refinement, L_X the tuple language of a typed symbol X, t_G the nonterminal thickness and θ_G the rule thickness ($t_G \le \theta_G$). ψ is the Parikh map and $\Psi(\vec{u}) = \sum_i \psi(u_i)$. For a typed non-start symbol X let $[X]$ be the $\equiv_{L,h}$-class containing L_X and

$$\kappa_G(X) = \min\{\|\vec{v}\|_1 : \vec{v} \in [X]\}, \qquad \kappa_G = \max_X \kappa_G(X).$$

The representative $\vec{v}$ need not belong to L_X, nor to any tuple language of G; it only has to have the type of X and the same distribution in L over *all* sentence contexts.

Conjecture 1 (E)**.** For fixed Σ, h, r there is a polynomial p such that every reduced good G of fan-out ≤ 2 and rank $\le r$ with $L(G) \in \mathcal{C}^{\mathrm{mcf}}_{2,h}$ satisfies $\kappa_G \le p(\|G\| \max\{1, \theta_G\})$.

2 Adversarial audit of R and D

2.1 Lemma R (relaxed witnesses)

Statement audited. For every non-start typed $X = A_{\mathbf{t}}$ pick $\omega'(X) \in [X]$; let $\chi'(X) = \Box_1$ if $\widetilde{S} \to X$ is a rule of $\widetilde{G}_0$, and otherwise any derivational sentence context of the untyped A in G. The set $K^\star$ of the words $\chi'(X)[\omega'(X)]$, $\chi'(X)[\rho(\omega'(Y_1), \ldots, \omega'(Y_q))]$ (typed rules $X \to \rho(Y_1, \ldots, Y_q)$, $q \geq 1$) and $\chi'(X)[\vec{c}]$ (typed constant rules) is characteristic for $\mathcal{B}^{\leq r}_{f,h}$ and $\mathcal{B}_{f,h}$.

Checks performed.

(R1) $[X]$ *is one class.* Tuples of L_X have type $\mathbf{t}$ and share a derivational typed context, so (f,h)-substitutability (arity $\mathrm{fan}(A) \leq f$) puts them in one $\equiv_{L,h}$-class.

(R2) *Membership.* A derivational context of A accepts every tuple of $L_A(G) \supseteq L_X$, hence every tuple of $[X]$; it lies in $\mathcal{E}_{\mathrm{fan}(A)}$ because G is good (v79, Lemma *Induced child context* iterated along the path). For a start-linked X the context $\Box_1$ accepts $L_X \subseteq L$, hence all of $[X]$. Rule words: $\omega'(Y_j) \equiv_{L,h} \vec{u}_j \in L_{Y_j}$, so v79, Lemma *Composition preserves equivalence*, gives $\rho(\omega'(Y_1), \ldots) \equiv_{L,h} \rho(\vec{u}_1, \ldots) \in L_X$; the latter is accepted by $\chi'(X)$. Constant rules produce members of L_X.

(R3) *Link rules.* $\omega'(X)$ and the rule tuple $\vec{z}_R = \rho(\omega'(Y_1), \ldots)$ both have type $\mathbf{t}$ (out_ρ of the child types) and share the observed context $\chi'(X)$, so Rule (Link) $[\omega'(X)] \to [\vec{z}_R]$ is generated.

(R4) *Composition witnesses.* $\chi'(X) \in \mathcal{E}_d$, child components are nonempty, and goodness (nonpermuting, nonmerging) separates consecutive components of one child either by terminal material, by a nonempty sibling component, or by an internal separator of $\chi'(X)$; hence each $\omega'(Y_j)$ is an observed tuple occurrence and the segmentation is a composition witness of rank $q \leq r$.

(R5) *Start, rank zero, ε, productivity.* Start-linked symbols use $\widetilde{S} \to [\omega'(X)]$ with $\omega'(X) \in K^\star$. Rank zero is covered by the third family of words. ε is outside good languages and is handled separately by the operator. $\widetilde{G}_0$ is trimmed, so every X is productive and reachable. Repeated words are harmless (sets).

(R6) *Interaction with v79 rule-only witnesses.* The canonical v79 Proposition *Rule-only characteristic witnesses* uses the identity $\omega(X) = \rho_X(\omega(Y_1), \ldots)$ for minimum-weight derivations. This identity is *false* for relaxed representatives, so the nonterminal words $\chi'(X)[\omega'(X)]$ must be kept; the cardinality bound of R is $|V||M|^f + |P||M|^{fr}$, not $|\mathcal{R}|$. The two optimisations must not be combined.

Verdict. R holds as stated; the completeness proof of v79 is used verbatim with $\iota(X) := [\omega'(X)]$, and soundness needs only $K^\star \subseteq L$.

2.2 Theorem D (abelian-filtered targets)

Statement audited. For $L = h^{-1}(\mathsf{Acc}) \cap \varphi^{-1}(\gamma + \mathcal{H})$ (any abelian group $\mathbb{A}$, homomorphism φ, subgroup $\mathcal{H}$) with a reduced good presentation G of fan-out $\leq f$ and rank $\leq r$, $\kappa_G \leq \mathrm{poly}(T)$ with $T = (r+1)t_G + \|G\|$, and hence a characteristic sample of total length $C(\|G\| \max\{1, \theta_G\})^e$, with C, e depending only on Σ, h, f, r.

Checks performed.

(D1) *Invariance (Step 1)* needs only a derivational context of each untyped A; reducedness supplies one. Uses nondeletion and linearity.

(D2) *Short defects (Steps 2–3)*: $\delta_R \in \Lambda$ because ω_A and $\rho(\omega_{B_1}, \ldots)$ are two yields of A; $\|\delta_R\|_1 \leq T$. The decomposition $\Psi(\vec{u}) - \Psi(\omega_A) = -\sum_\nu \delta_{R_\nu}$ over the nodes of any derivation is exact (rank-zero nodes included).

(D3) *Integer program (Step 4).* Unknowns: the coefficients of the periods of fixed semilinear representations of $\psi(h^{-1}(m)\cap\Sigma^+)$ (at most fC_h) and $\mu_\lambda^\pm$ for $\lambda\in\Gamma_T$ (at most $2(2T+1)^{|\Sigma|}$). Equations: $|\Sigma|$. Entries of the matrix and of the right-hand side $\Psi(\omega_A)-\sum_i c_i$ are bounded by $T+fC_h$. The system is solvable (by any $\vec{u}\in L_X$), so the small-solution theorem yields a solution with entries $\le n(ma)^{2m+1}$, $m=|\Sigma|$, which is poly(T) with exponent $O(|\Sigma|)$.

(D4) *Uniformity.* Γ_T depends on Λ, but only its cardinality enters the bound, and $|\Gamma_T|\le(2T+1)^{|\Sigma|}$ for *every* Λ. The constants therefore do not depend on $\mathbb{A}$, φ, $\mathcal{H}$, γ or Acc (in particular not on the index of $\mathcal{H}$).

(D5) *Equivalence (Step 5)* uses commutativity of $\mathbb{A}$ only through $\varphi=\bar{\varphi}\circ\psi$.

(D6) ε. If $\varepsilon\in h^{-1}(\mathsf{Acc})\cap\varphi^{-1}(\gamma+\mathcal{H})$ the good presentation generates $L\setminus\{\varepsilon\}$; Step 5 only evaluates nonempty words $E[\vec{v}]$, so nothing changes.

(D7) *External sources.* The small-solution theorem is Papadimitriou, *On the complexity of integer programming*, J. ACM 28 (1981) 765–768, and MIT/LCS/TM-152 (1980). The primary PDF could not be downloaded from this environment (repository returned an access error); the bound $n(ma)^{2m+1}$ was checked against its restatement in Fomin, Panolan, Ramanujan, Saurabh, arXiv:1607.05342, Sec. 1. Parikh's theorem is applied to the regular sets $h^{-1}(m)$, where it is elementary.

Verdict. D holds as stated, uniformly in the abelian data. The only external dependency not checked in a primary source is the small-solution bound, which is standard and is restated in the cited arXiv paper.

3 Parikh rigidity of intra-word exchanges

Lemma 2 (P)**.** *Let $E\in\mathcal{E}_d$ and $\vec{x},\vec{y}\in(\Sigma^+)^d$ with $E[\vec{x}]=E[\vec{y}]$. Then $\Psi(\vec{x})=\Psi(\vec{y})$; in particular $\|\vec{x}\|_1=\|\vec{y}\|_1$.*

Proof. $\psi(E[\vec{x}])=\psi(E)+\Psi(\vec{x})$ where $\psi(E)$ counts the fixed material, and likewise for $\vec{y}$. □

Corollary 3. *Every exchange pair produced by the power-transfer lemma (B) or the local shift-pair lemma (B′) of v79 is Parikh-equal. Consequently no sequence of such exchanges, applied in any contexts, changes the Parikh vector of a word or of a tuple; local mobility can never by itself produce a shorter member of a distribution class. A shorter representative $\vec{y}\equiv_{L,h}\vec{x}$ with $\|\vec{y}\|_1<\|\vec{x}\|_1$ requires a shared accepting context E with $E[\vec{x}]\neq E[\vec{y}]$, i.e.* two different words *of L.*

This answers the methodological question of v79: Ramsey-type shift pairs are a source of *closure* constraints (they force languages to be closed under block moves), not of *shortening*. Every shortening argument proved so far (D and F1-sl below) instead combines a closure constraint with the grammar: the grammar supplies short yields of untyped nonterminals, and the closure constraint lets them be converted into short representatives of the correct type.

4 Semilattice-split abelian unions (F1-sl)

Fix a nonempty set $\Sigma_{\mathsf{c}}\subseteq\Sigma$ of *marker* letters. For a word w let $\mathsf{c}(w)=1$ if w contains a marker and $\mathsf{c}(w)=0$ otherwise; for a tuple $\mathsf{c}(\vec{u})=\max_i\mathsf{c}(u_i)$ and for a context $\mathsf{c}(E)$ is 1 iff its fixed material contains a marker. The typing $h_{\mathrm{U}}:\Sigma^*\to(\{0,1\},\max)$, $w\mapsto\mathsf{c}(w)$, is the two-element semilattice typing; it is not in **LG**.

Definition 4. Let $\mathbb{A}$ be an abelian group, $\varphi : \Sigma^* \to \mathbb{A}$ a homomorphism, $\mathcal{H} \leq \mathbb{A}$, and $\gamma_0, \gamma_1 \in \mathbb{A}$. Put

$$L = \{w \in \Sigma^+ : \varphi(w) \in \gamma_{\mathsf{c}(w)} + \mathcal{H}\}.$$

We call L a *semilattice-split abelian union* and write $g := \gamma_1 - \gamma_0$, $\Lambda := \{v \in \mathbb{Z}^\Sigma : \bar{\varphi}(v) \in \mathcal{H}\}$ where $\varphi = \bar{\varphi} \circ \psi$.

If $g \in \mathcal{H}$ then $L = \Sigma^+ \cap \varphi^{-1}(\gamma_0 + \mathcal{H})$ is an abelian filter and Theorem D applies. If $g \notin \mathcal{H}$, L is in general *not* of the form $h^{-1}(\mathsf{Acc}) \cap \varphi'^{-1}(\gamma + \mathcal{H}')$: for instance $\Sigma = \{\mathsf{a}, \mathsf{c}\}$, $\Sigma_\mathsf{c} = \{\mathsf{c}\}$, $\varphi(w) = |w|_\mathsf{a} \in \mathbb{Z}/N\mathbb{Z}$, $\gamma_0 = 0$, $\gamma_1 = 1$ gives the offset that would have to be a monoid map $\{0,1\} \to \mathbb{Z}/N\mathbb{Z}$ with $1 \mapsto 1$, which is impossible since $1 + 1 \neq 1$.

Lemma 5 (Membership)**.** *Let h be a finite typing with $h_\mathrm{U} \preceq h$ and $f \geq 1$. A semilattice-split abelian union is (f,h)-tuple-substitutable. Hence it belongs to $\mathcal{C}^{\mathrm{mcf}}_{f,h}$ whenever it is an f-MCFL.*

Proof. Let $\vec{x}, \vec{y}$ have equal h-types and share an accepting context E_0. Equal h-types give equal h_U-types componentwise, so $\mathsf{c}(E[\vec{x}]) = \mathsf{c}(E[\vec{y}])$ for every context E. From E_0: $\varphi(\vec{x}) - \varphi(\vec{y}) \in \mathcal{H}$ (both fillings lie in the same branch $\gamma_\beta + \mathcal{H}$). For any E the two fillings are in the same branch and their φ-values differ by an element of $\mathcal{H}$, so they are accepted simultaneously. □

From now on $g \notin \mathcal{H}$, and $G = (V, \Sigma, P, S)$ is a reduced good presentation of L of fan-out $\leq f$ and rank $\leq r$. A *derivational context* of $A \in V$ is a sentence context obtained from a successful derivation by deleting one A-subtree; it accepts every tuple of $L_A := L_A(G)$. Call A *bi-typed* if L_A contains tuples $\vec{u}$ with $\mathsf{c}(\vec{u}) = 0$ and with $\mathsf{c}(\vec{u}) = 1$, *marker-only* if $\mathsf{c} \equiv 1$ on L_A, *marker-free* if $\mathsf{c} \equiv 0$ on L_A, and *open* if it has a derivational context E with $\mathsf{c}(E) = 0$.

Lemma 6 (Offsets)**.** *Let E be a derivational context of A and $\vec{u}, \vec{u}' \in L_A$.*

(a) If $\mathsf{c}(E) = 1$ then $\varphi(\vec{u}) - \varphi(\vec{u}') \in \mathcal{H}$.

(b) If $\mathsf{c}(E) = 0$ then $\varphi(\vec{u}) - \varphi(\vec{u}') \in (\mathsf{c}(\vec{u}) - \mathsf{c}(\vec{u}'))g + \mathcal{H}$.

(c) A bi-typed open nonterminal has no derivational context with $\mathsf{c}(E) = 1$.

(d) If A is not both bi-typed and open, then $\Psi(\vec{u}) - \Psi(\vec{u}') \in \Lambda$ for all $\vec{u}, \vec{u}' \in L_A$. If A is bi-typed and open, this holds whenever $\mathsf{c}(\vec{u}) = \mathsf{c}(\vec{u}')$.

Proof. $E[\vec{u}] \in L$ lies in branch $\mathsf{c}(E) \vee \mathsf{c}(\vec{u})$, so $\varphi(E) + \varphi(\vec{u}) \in \gamma_{\mathsf{c}(E) \vee \mathsf{c}(\vec{u})} + \mathcal{H}$. Subtracting the same relation for $\vec{u}'$ gives (a) and (b). For (c), take $\vec{u}_0, \vec{u}_1$ with $\mathsf{c} = 0, 1$: (a) and (b) would give $\varphi(\vec{u}_1) - \varphi(\vec{u}_0) \in \mathcal{H} \cap (g + \mathcal{H})$, forcing $g \in \mathcal{H}$. For (d): every A has a derivational context (reducedness). If A is not bi-typed, c is constant on L_A and (a) or (b) gives the claim. If A is bi-typed and not open, all its derivational contexts have $\mathsf{c}(E) = 1$ and (a) applies. If A is bi-typed and open, (b) with $\mathsf{c}(\vec{u}) = \mathsf{c}(\vec{u}')$ applies. □

Let V_1 be the set of bi-typed open nonterminals.

Lemma 7 (Offset rigidity)**.** *Let $A \in V_1$ and $R : A \to \rho(B_1, \ldots, B_q)$ a rule whose template writes no marker and none of whose children is marker-only. Then at most one B_k is bi-typed, and every bi-typed B_k belongs to V_1.*

Proof. Let E_A be an open derivational context of A. Suppose B_{k_1} and B_{k_2} ($k_1 \neq k_2$) are bi-typed. Choose $\vec{x}_0, \vec{x}_1 \in L_{B_{k_1}}$ and $\vec{y}_0, \vec{y}_1 \in L_{B_{k_2}}$ with c-values $0, 1$, and for every other k a tuple of L_{B_k} with $\mathsf{c} = 0$ (it exists because B_k is not marker-only). Let $\vec{z}_{ab}$ be the yield of A obtained by applying ρ to these choices with $\vec{x}_a, \vec{y}_b$. Since ρ writes no marker, $\mathsf{c}(\vec{z}_{ab}) = a \vee b$. By Lemma 6(b) for E_A,

$$\varphi(\vec{x}_1) - \varphi(\vec{x}_0) = \varphi(\vec{z}_{10}) - \varphi(\vec{z}_{00}) \in g + \mathcal{H}, \qquad \varphi(\vec{x}_1) - \varphi(\vec{x}_0) = \varphi(\vec{z}_{11}) - \varphi(\vec{z}_{01}) \in \mathcal{H},$$

so $g \in \mathcal{H}$, a contradiction. For the second claim, filling the siblings of a bi-typed B_k with marker-free tuples inside E_A and ρ yields a derivational context of B_k (it comes from a successful derivation) without markers; it lies in $\mathcal{E}_{\mathrm{fan}(B_k)}$ by goodness. Hence B_k is open. □

Lemma 8 (Short canonical yields)**.** *Put $T_0 := |V|\,(\|G\| + r\,t_G)$. Every $A \in V$ has a yield ω_A with $\|\omega_A\|_1 \le t_G$. Every $A \in V_1$ has yields ω_A^0, ω_A^1 with $\mathsf{c}(\omega_A^b) = b$ and $\|\omega_A^b\|_1 \le T_0$.*

Proof. The first claim is the definition of t_G. For $b \in \{0,1\}$ and $A \in V_1$ let $\lambda_b(A)$ be the minimum of $\|\vec{u}\|_1$ over $\vec{u} \in L_A$ with $\mathsf{c}(\vec{u}) = b$, and choose a derivation realising it that is lexicographically minimal in (yield length, number of nodes). Let $R : A \to \rho(B_1, \dots, B_q)$ be its root rule. By optimal substructure, a child subderivation whose c-value can be changed without changing c of the root is a shortest yield of that child.

$b = 0$. The template writes no marker and all child yields have $\mathsf{c} = 0$, so no child is marker-only. By Lemma 6 and Lemma 7 at most one child B is bi-typed, and it lies in V_1. Every other child is marker-free, its subyield is a shortest yield ($\le t_G$), and B's subyield realises $\lambda_0(B)$. Hence $\lambda_0(A) \le \|G\| + (r-1)t_G + \lambda_0(B)$, or $\le \|G\| + rt_G$ if no child is bi-typed.

$b = 1$. If the template writes a marker or some child is marker-only, every child subyield may be replaced by a shortest yield without losing the marker, so $\lambda_1(A) \le \|G\| + rt_G$. Otherwise Lemma 7 applies: at most one child is bi-typed, the others are marker-free, so the marker comes from the unique bi-typed child $B \in V_1$, whose subyield realises $\lambda_1(B)$; thus $\lambda_1(A) \le \|G\| + (r-1)t_G + \lambda_1(B)$.

In both cases the passage $A \mapsto B$ strictly decreases the pair (length, node count) of the chosen minimal derivation, so a chain visits each nonterminal at most once and $\lambda_b(A) \le |V|(\|G\| + rt_G)$. □

For $A \in V$ and $b \in \{0,1\}$ put $\omega_A^{(b)} := \omega_A^b$ if $A \in V_1$ and $\omega_A^{(b)} := \omega_A$ otherwise.

Lemma 9 (Short Λ-defects)**.** *Let $T_1 := (r+1)T_0 + \|G\|$, $\Gamma := \{\lambda \in \Lambda : \|\lambda\|_1 \le T_1\}$ and let Λ_1 be the subgroup generated by Γ. For every $A \in V$ and $\vec{u} \in L_A$,*

$$\Psi(\vec{u}) - \Psi(\omega_A^{(\mathsf{c}(\vec{u}))}) \in \Lambda_1.$$

Proof. For a rule $R : A \to \rho(B_1, \dots, B_q)$ and $\vec{b} \in \{0,1\}^q$ with each b_k realised by L_{B_k}, let $\vec{z} = \rho(\omega_{B_1}^{(b_1)}, \dots, \omega_{B_q}^{(b_q)}) \in L_A$ and let $\vec{u}'$ be any yield of A obtained by applying R to children $\vec{u}_k$ with $\mathsf{c}(\vec{u}_k) = b_k$. *Claim:* if $A \in V_1$ then $\mathsf{c}(\vec{z}) = \mathsf{c}(\vec{u}')$. If ρ writes a marker or some child is marker-only, both values are 1. Otherwise Lemma 7 shows that every bi-typed child lies in V_1, so for every child $\mathsf{c}(\omega_{B_k}^{(b_k)}) = b_k$ (children not in V_1 are then marker-free and $b_k = 0$), and both values equal $\max_k b_k$.

Put $\beta := \mathsf{c}(\vec{u}')$ and $\delta_{R,\vec{b}} := \Psi(\omega_A^{(\beta)}) - \Psi(\vec{z})$. If $A \in V_1$, $\omega_A^{(\beta)}$ and $\vec{z}$ are yields of A with the same c-value by the claim; if $A \notin V_1$, the first part of Lemma 6(d) applies. In both cases $\delta_{R,\vec{b}} \in \Lambda$, and $\|\delta_{R,\vec{b}}\|_1 \le T_0 + \|G\| + rT_0 = T_1$.

Now induct on a derivation of $\vec{u} \in L_A$ with root rule R and child yields $\vec{u}_k$; put $b_k = \mathsf{c}(\vec{u}_k)$, so $\vec{u}' = \vec{u}$ and $\beta = \mathsf{c}(\vec{u})$. Then

$$\Psi(\vec{u}) - \Psi(\omega_A^{(\mathsf{c}(\vec{u}))}) = \sum_k \big(\Psi(\vec{u}_k) - \Psi(\omega_{B_k}^{(b_k)})\big) - \delta_{R,\vec{b}} \in \Lambda_1,$$

using the induction hypothesis for the children (rank-zero rules are the base case, with empty sum). □

Theorem 10 (F1-sl)**.** *Fix Σ, Σ_{c}, a finite typing $h : \Sigma^* \to M$ with $h_{\mathrm{U}} \preceq h$, $f \ge 1$ and $r \ge 1$. There are C, e depending only on (Σ, h, f, r) such that for every semilattice-split abelian union L*

(arbitrary $\mathbb{A}, \varphi, \mathcal{H}, \gamma_0, \gamma_1$) and every reduced good presentation G of L of fan-out $\le f$ and rank $\le r$:

$$\kappa_G \le C\big(\|G\| \max\{1, \theta_G\}\big)^e,$$

and $\mathcal{B}_{f,h}^{\le r}$ has a characteristic sample $K^\star \subseteq L$ with $|K^\star| \le |V||M|^f + |P||M|^{fr}$ and $\|K^\star\|_+ \le C'(\|G\| \max\{1, \theta_G\})^{e'}$. In particular $\mu_m(L)$ is polynomially bounded for every $m \in h(L)$.

Proof. If $g \in \mathcal{H}$ apply Theorem D. Otherwise let $X = A_{\mathbf{t}}$ be a non-start typed symbol of arity d, fix $\vec{u} \in L_X$, and put $\beta := \mathsf{c}(\vec{u})$, which is determined by $\mathbf{t}$ because $h_{\mathrm{U}} \preceq h$. Fix semilinear representations of $\psi(h^{-1}(m) \cap \Sigma^+)$, $m \in M$, with at most C_h linear sets and periods and constants of norm $\le C_h$. The integer program asking for $\vec{v}$ with $h(v_i) = t_i$ and $\Psi(\vec{v}) - \Psi(\omega_A^{(\beta)}) \in \Lambda_1$, written with the period coefficients of the chosen linear sets and coefficients $\mu_\lambda^{\pm}$ ($\lambda \in \Gamma$), has $|\Sigma|$ equations, at most $fC_h + 2(2T_1+1)^{|\Sigma|}$ unknowns and entries bounded by $T_1 + fC_h$. It is solvable by Lemma 9 (witness $\vec{u}$), hence has a solution with entries $\mathrm{poly}(T_1)$, giving $\vec{v}$ with $h^{(d)}(\vec{v}) = \mathbf{t}$ and $\|\vec{v}\|_1 \le \mathrm{poly}(T_1)$.

$\vec{v} \equiv_{L,h} \vec{u}$: both have type $\mathbf{t}$, so for every context E the fillings lie in the same branch $\mathsf{c}(E) \vee \beta$; and $\Psi(\vec{v}) - \Psi(\vec{u}) \in \Lambda_1 \subseteq \Lambda$, so their φ-values differ by an element of $\mathcal{H}$. This is the step that uses the *common* subgroup $\mathcal{H}$ of both branches. Hence $\kappa_G(X) \le \mathrm{poly}(T_1)$; since $t_G \le \theta_G$, $T_1 \le (r+1)|V|(\|G\| + r\theta_G) + \|G\|$. The sample bounds follow from the relaxed-witness lemma R and its corollary; the μ_m bound from the type-deletion proposition of v79. □

Remark 11 (What is new, and what is not). (i) $h_{\mathrm{U}} \notin \mathbf{LG}$, so the local-group theorem of v79 does not apply, and for $g \notin \mathcal{H}$ the target is not an abelian filter, so Theorem D does not apply directly. (ii) The proof does not use any substitutability property of L beyond Lemma 5; the decisive input is *offset rigidity* (Lemma 7), which is a property of every presentation of L. (iii) The doubling grammar $A_i \to A_{i-1}A_{i-1}$ cannot occur on a marker-free path below an open bi-typed nonterminal; this is exactly what prevents exponentially rare marker-free words with small ordinary thickness. (iv) The theorem is stated with one subgroup $\mathcal{H}$ for both branches. The exact scan `v80_branch_union_scan.py` (cyclic $\mathbb{A} = \mathbb{Z}/k\mathbb{Z}$, $k \le 6$, alphabet $\{\mathsf{a}, \mathsf{c}\}$, marker c, all pairs of value sets) found 1 364 distinct two-branch unions in $\mathcal{C}_{2,h_{\mathrm{U}}}^{\mathrm{mcf}}$ that are not single abelian filters; *all* 1 364 become common-subgroup unions after intersecting with the values realisable in each branch. This supports Conjecture 12 below but proves nothing beyond $k \le 6$.

5 Further evidence and open lemmas

5.1 Boolean combinations under a content typing (finite scan)

`v80_content_union_scan.py` takes $\Sigma = \{\mathsf{a}, \mathsf{c}, \mathsf{d}\}$, the content typing $h(w) = \{\text{markers in } w\} \subseteq \{\mathsf{c}, \mathsf{d}\}$ (a semilattice, not in $\mathbf{LG}$), $\varphi : \Sigma^* \to \mathbb{Z}/k\mathbb{Z}$ arbitrary, and *every* relation $R \subseteq \mathcal{P}(\{\mathsf{c}, \mathsf{d}\}) \times \mathbb{Z}/k\mathbb{Z}$, with $L = \{w \in \Sigma^+ : (h(w), \varphi(w)) \in R\}$. For $k \le 3$ it decides $(2,h)$-tuple-substitutability exactly (the monoid is commutative, so tuples and contexts reduce to sums). Of 81 456 relations, 6 958 give class members. Of these, 6 894 are *twisted common-subgroup filters* ($R_S = (g_S + \mathcal{H}) \cap W_S$ for one $\mathcal{H}$ and offsets g_S), and the remaining 64 are *branchwise filters* with branch-dependent subgroups $\mathcal{H}_S$. No other shape occurs.

Conjecture 12 (F1-gen: shape)**.** If h is a content typing, φ maps to an abelian group and $L = \{w \in \Sigma^+ : (h(w), \varphi(w)) \in R\}$ lies in $\mathcal{C}_{2,h}^{\mathrm{mcf}}$, then for every realised content S the branch R_S is a coset of a subgroup $\mathcal{H}_S$ intersected with the realised values, and $\mathcal{H}_S$ and $\mathcal{H}_{S'}$ agree on the differences of tuples whose type pattern can reach both S and S'.

Conjecture 13 (F1-tw: data)**.** For fixed (Σ, h, f, r) with h a content typing, every target of the shape in Conjecture 12 with a reduced good presentation G satisfies $\kappa_G \le \mathrm{poly}(\|G\| \max\{1, \theta_G\})$.

Evidence. Theorem 10 is the case of one marker and a common subgroup. The proof idea extends as follows (not yet a proof). Offsets are now $o(T) = g_{S_E \cup T}$ for a context of content S_E. The rectangle identity from Lemma 7 forbids two children of an open node whose content variations overlap in a marker with nonzero offset increment, because $\cup$ is idempotent. Children whose variations are disjoint are allowed (offsets additive across distinct markers), so the "variation tree" of a minimal derivation has at most $|\Sigma_{\mathsf{c}}|$ leaves, which would bound the canonical yields by $|\Sigma_{\mathsf{c}}|\,|V|(\|G\| + rt_G)$. Making "variation" precise and handling branch-dependent subgroups are the missing steps.

5.2 Principles that fail

(F1) *"Local mobility implies shortening."* False as a method: by Lemma 2 every exchange witnessed inside one word is Parikh-equal (Corollary 3).

(F2) *"A branch-coherent union reduces to one filter."* False: for $g \notin \mathcal{H}$ a semilattice-split union is not an abelian filter (Section 4). The data bound survives for a different reason (offset rigidity).

(F3) *"Two-branch unions with different subgroups are covered by the same proof."* The proof of Theorem 10 uses the common subgroup in Step 5. In the scan every two-branch h_{U}-union turned out to admit a common subgroup after restriction to realised values, but under the two-marker content typing 64 members genuinely need branch-dependent subgroups; they are not covered.

5.3 Status of Conjecture E

Proved cases: $h(\Sigma^+) \in \mathbf{LG}$ (v79 local-group theorem, since $L_X \subseteq [X]$); abelian filters for every h (Theorem D); semilattice- split common-subgroup unions for every $h \succeq h_{\mathrm{U}}$ (Theorem 10). Disproved at fan-out one (companion CFG paper). Open in general. No counterexample candidate survived: every counterexample design reduces either to a rigid word (destroyed by the power-transfer and shift-pair lemmas), or to a type-commutative target (the abelian, split and conjectured F1-tw cases).

5.4 Next lemmas (precise)

(N1) *Branchwise subgroups.* Extend Theorem 10 to $L = \{w : \varphi(w) \in \gamma_{\mathsf{c}(w)} + \mathcal{H}_{\mathsf{c}(w)}\}$ with $\mathcal{H}_0 \cap \mathbb{Z}^{\Sigma\setminus\Sigma_{\mathsf{c}}}$ and $\mathcal{H}_1 \cap \mathbb{Z}^{\Sigma\setminus\Sigma_{\mathsf{c}}}$ (pulled back to Λ_0, Λ_1) equal.

(N2) *Variation tree.* For a content typing and an open nonterminal A, define $\mathrm{Var}(A) \subseteq \Sigma_{\mathsf{c}}$ as the set of markers c with $g_{S\cup\{\mathsf{c}\}} \neq g_S$ for some realisable content difference. Prove that in every rule the children with $\mathrm{Var} \neq \emptyset$ have pairwise disjoint Var.

(N3) *Type-commutative normal form.* If $L \in \mathcal{C}^{\mathrm{mcf}}_{2,h}$ and the marked part of L is a rich abelian filter (every coset met by marked contexts), then the marker-free part of L is determined by $(h(w), \varphi(w) + \mathcal{H})$; prove this and reduce to N1–N2.

(N4) *Beyond commutative.* Find a target in $\mathcal{C}^{\mathrm{mcf}}_{2,h}$ whose classes are not determined by type and an abelian invariant at large scale, or prove that none exists (which together with N1–N2 would give Conjecture E).